Non-additive Strain–Dopant Energetics in Quasi-Hexagonal C_{60} Graphullerene

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Under mechanically constrained loading on quasi-hexagonal (qHP) C_{60} graphullerene, measurable deviations from additive strain–dopant energetics arise because frozen elastic incompatibility introduces a bilinear strain–composition interaction that can reach magnitudes comparable to high-throughput screening margins—not because a finite-difference estimator is merely nonzero. The additive approximation in sequential workflows omits the discrete strain–substitution interaction for each fixed substituent. Matched four-corner audits on the 400 Ry core (n=1, 2, and 4; PBE+D3, affine biaxial strain with locked dopant sites) show that this term can alter energy differences at the scale relevant to energetic screening for size-mismatched phosphorus. The representative core value is $|\mathcal{S}| = 23.7\pm2$ meV/atom at n=4 ($\eta \approx 8\%$, about 12 times the 2 meV/atom reporting floor). Across the audited sizes, P spans 23.7–31.9 meV/atom; the n=1 value (31.9 ± 2 meV/atom) is the strongest constrained coupling and also the weakest Γ-only sampling support. Supercells with $n \geq 6$ (350 Ry) are reference-only and are not used for that screening comparison. Cluster tetramer "strain" grids do not affine-scale the molecular coordinates and are not used as strain-physics evidence. The four-corner difference $\mathcal{S}$ estimates that coupling on the audited grid and is not a new thermodynamic observable. It is not a charged-defect or polaron-rate estimator. The n=2 $\mathcal{S}(n)$ row is reference-limited at the undoped corner, not a sign change of the discrete species interaction. On the 400 Ry core, only the size-mismatched substituent is numerically resolved well above the protocol floor: $|\mathcal{S}|$ is 12–16 times the 2 meV/atom reporting band for P (31.9 meV/atom at n=1; 23.7 meV/atom at n=4), while N at n=1 (+2.30 meV/atom) sits at that floor and is not assigned separate physical significance. The pattern is consistent with covalent-radius mismatch on this B/N/P map; it is not a unique causal assignment. Ionic relaxation of periodic n=1 P suppresses the constrained cross term to the reporting floor ($|\mathcal{S}_{\rm relaxed}| < 10^{-3}$ meV/atom). On a separate tetramer boundary condition the same relaxation reverses the sign and increases $|\mathcal{S}|$ (+1.19 to −2.28 meV/atom). Rigid $|\mathcal{S}|$ is an upper-end protocol estimate relative to the fully relaxed fixed-cell protocol studied here, not a universal bound on every constraint manifold. The audit signals when sequential additive workflows should be replaced by explicit coupled evaluations on the $E(\epsilon, \delta)$ landscape.

I. INTRODUCTION

Substitutional doping and mechanical strain are widely used to tune stability and electronic structure in covalent molecular networks [1, 2]. Sequential screens often add separate strain-only and substitution-only increments, omitting the discrete strain–substitution interaction for each fixed dopant species [1, 3, 4]. Cluster-expansion Hamiltonians already include composition–strain cross terms through constituent-strain and configurational interactions [5, 6]; what has been missing for soft fullerene networks is a matched four-corner total-energy audit under simultaneous load. Expanding the energy per atom $e(\epsilon, x) = E(\epsilon, \delta)/N_{\rm atoms}$ about the pristine reference gives

$$e(\epsilon, x) \approx e_0 + a\,\epsilon + b\,x + c\,\epsilon\,x + \mathcal{O}(\epsilon^2, x^2, \epsilon^2 x, \epsilon x^2), \quad (1)$$

where $c = \partial^2 e/\partial\epsilon\,\partial x$ is the lowest-order bilinear coupling. Here x is a local continuous composition coordinate introduced at one fixed species; B, N, and P are not three points on one shared x axis. The four-corner estimator $\mathcal{S}_\delta(\epsilon)$ (Sec. II C) is the discrete strain–substitution interaction at +3% biaxial load. Only after that local coordinate is introduced does $\mathcal{S}_\delta(\epsilon)$ sample $\partial^2 e/\partial\epsilon\,\partial x$. It is a diagnostic, not a new thermodynamic observable (Supplemental Material, Sec. II) [7]. It is not a charged-defect or polaron-rate estimator.

Finite-size $\mathcal{S}(n)$ entries that fail the undoped-corner SCF gate are reference-limited and must not be read as a dopant-rank trend.

We study quasi-hexagonal (qHP) C_{60} graphullerene [8–10] as an elastically soft, experimentally accessible host. Stress-related anisotropy of qHP versus qTP C_{60} has been mapped for optoelectronic device implications [11]; that comparison is a uniaxial-stress geometry map, not a four-corner $\mathcal{S}$ on a substitutional B/N/P grid. Substitutional B, N, and P form a donor, acceptor, and size-mismatch triad spanning both signs of covalent-radius mismatch relative to C ($\Delta r_{\rm cov}$ = +7, −6, and +30 pm; Table V). Prior DFT work maps strain and doping on separate grids or reports nonlinear electronic response without matched four-corner *total-energy* audits [3, 4, 9, 12–16]; a positioning table is in Supplemental Material, Table I. A related strain-only map of polymerized C_{24} graphullerene [17] is chemically distinct from the qHP C_{60} lattice used here and likewise omits that four-corner $\mathcal{S}$ audit. Atomic-scale mechanical and tribological maps of graphullerene [18] likewise do not evaluate a four-corner total-energy $\mathcal{S}$. A substitutional defect on a pre-strained cage, together with global biaxial load, can store elastic incompatibility that linear α does

not capture. Here we test whether additive totals fail under joint (ϵ, δ) load, whether the omitted term matters under mechanically constrained configurations, and whether failure tracks covalent-radius mismatch rather than π-electron descriptors alone. The protocol is reported for one host; transferring it requires repeating the same (ϵ, δ, n) grid.

II. COMPUTATIONAL METHODS

A. Electronic structure method

All total-energy calculations use CP2K [19] with Goedecker–Teter–Hutter (GTH) pseudopotentials, the DZVP-MOLOPT-SR-GTH atom-centered basis, and the Gaussian and plane waves (GPW) scheme.

a. Exchange-correlation and dispersion. We use PBE [20] with Grimme DFT-D3 (Becke–Johnson damping) [21] for the production dataset. Hybrids or meta-GGAs (SCAN, r^{2}SCAN) can shift absolute totals by tens of meV per atom in carbon networks; production rankings are therefore framed as *within-PBE+D3* comparisons of $\mathcal{S}$ among B, N, and P under matched load. The four-corner construction [Sec. II C] subtracts totals computed with identical XC, basis, dispersion, and numerical settings, so systematic shifts largely cancel when *ranking* dopants at fixed protocol—but cancellation is not assumed without audit: on the isolated-cage tetramer P grid (not a strained covalent network), Grimme-D3 non-additivity is $\mathcal{S}_{\mathrm{vdW}} \approx +0.33$ meV/atom (Table I)—a cluster-protocol bound, not a strain-physics decomposition of periodic $\mathcal{S}$. No meta-GGA $\mathcal{S}$ values are reported (Table I; Limitations).

b. Plane-wave mesh and Brillouin-zone sampling. Plane-wave cutoffs are 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (REL_CUTOFF 50 Ry; QS EPS_DEFAULT 10^{-10} E_{h}). Comparisons at fixed n always use the same cutoff. Production supercells contain 60–480 atoms with in-plane lattice constants up to ∼44 Å at n=8; inputs use Γ-only sampling. Because $\mathcal{S}$ is a second-order difference on the *same* supercell, uniform k-point or cutoff shifts that co-drifts all four corners largely cancel in the four-corner estimator; sensitivity must be audited on $\mathcal{S}$(cutoff) and $\mathcal{S}(k)$, not on bare totals alone. A partial cutoff audit at $6 \times \mathrm{C}_{60}$ N and +3% shifts one corner by only ∼0.057 meV/atom between 350 and 400 Ry, implying $|\Delta\mathcal{S}| < 0.2$ meV/atom when only the mesh drifts (Table I). Dense-k $(2 \times 2 \times 1)$ four-corner benchmarks on n=1 B/N/P at +3% are not included in the production set (Table I; Limitations); the smallest reported $|\mathcal{S}|$ values (2.3–3.0 meV/atom for N and B at n=1) are therefore interpreted with explicit Γ-only caution.

c. SCF convergence and four-corner precision. Unless noted otherwise, the SCF threshold is EPS_SCF = 10^{-6} E_{h} in inner and outer orbital-transformation loops (MAX_SCF 300; OUTER_SCF MAX 30). The $2 \times \mathrm{C}_{60}$ cells are the exception: they cycled at 10^{-6} and were converged at 10^{-5} E_{h}, which is why their pristine reference lies off the size-converged plateau and why we flag that row as reference-limited (Table I; Sec. III B). Because the four-corner stencil of Eq. (3) includes that pristine total, the n=2 row is reference-limited at the undoped corner rather than by a sign change of the discrete species interaction. The n=8 row is likewise reference-limited at the undoped corner: the undoped +3% strain cost sits well above the n=1/4/6 plateau even though that cell uses the same 350 Ry production cutoff as n=6, so the offset is not a cutoff-class change and is not reported as a size-trend sign change. Charged-state workflows in the Supplemental Material tighten this to 10^{-7} E_{h}; selected stiff tetramer corners may use 10^{-5} E_{h} when outer SCF cycling persists at 10^{-6}, which stays inside the ∼2 meV/atom reporting floor of Table I. Because $\mathcal{S}$ is built from four independently converged totals [Eq. (2)], uncertainty is propagated as $\sigma_{\mathcal{S}} = 2\sigma_e$ for uncorrelated per-atom corner errors σ_e [Eq. (5); Supplemental Material, Sec. II E] (Table I). We use $\sigma_{\mathcal{S}} \lesssim 2$ meV/atom as the conservative reporting band on the $n \leq 4$ production grid.

d. Spin, charge, and mechanical boundary conditions. Substitutional periodic cells are charge-neutral closed-shell Kohn–Sham calculations (no spin polarization in the production $\mathcal{S}$ grid); Hirshfeld strain-path audits on n=1 B/N/P use spin-polarized KS only where odd electron counts require it (Sec. II D 3). We do not apply compensating background charge or charged supercells in the $\mathcal{S}$ audit: doping is modeled as neutral substitution.

B. Strain and doping models

Biaxial strain is $\epsilon = (a - a_0)/a_0 \times 100\%$, with a (a_0) the strained (equilibrium) in-plane lattice constant; $\delta \in \{\mathrm{B}, \mathrm{N}, \mathrm{P}\}$ labels the substituent and is not a continuous coordinate shared by B, N, and P. $\mathcal{S}$ is defined in Sec. II C; π-projected densities of states are written π-DOS. Periodic $n \times \mathrm{C}_{60}$ cells place one substituent per cage (∼1.67 at.%) and affine-scale both lattice vectors and Cartesian coordinates; that protocol supplies every $\mathcal{S}(n)$ value in this paper. Isolated-cage tetramer clusters ($4 \times \mathrm{C}_{60}$, ∼5 at.% substitution) are not a covalently networked monolayer: labeled tetramer "strain" grids change only the vacuum box and do not affine-scale the molecular coordinates, so they are not used as strain-physics evidence. The substitution energy $\Delta E_{\mathrm{sub}} = (E_{\mathrm{doped}} - E_{\mathrm{pristine}})/n_{\mathrm{dopants}}$ is not a formation enthalpy (no elemental chemical potentials).

a. Rigid vs. relaxed protocols. Rigid strain on periodic cells scales lattice vectors at fixed fractional coordinates and is the production protocol for $\mathcal{S}(n)$. It is a constrained fixed-coordinate diagnostic for loads in which ions cannot relax (epitaxial clamp, strong adhesion, or ultrafast load): an upper-end protocol estimate relative to the fully relaxed fixed-cell protocol studied here, not a universal bound on every con-

straint manifold and not an equilibrium free-energy difference. *Relaxed strain* uses fixed-cell BFGS at each bilinear corner on the periodic n=1 P cell before evaluating $\mathcal{S}$ [Fig. 2(c)]. Force thresholds are 4.5×10^{-4} and 3.0×10^{-4} Ha/bohr ($\sim$0.023 and $\sim$0.015 eV/Å) at ϵ=0 and +3%, tighter than typical 0.02 eV/Å defaults so that near-zero relaxed $|\mathcal{S}|$ is resolved. On that periodic cell, relaxation suppresses the constrained cross term ($|\mathcal{S}_{\mathrm{relaxed}}| < 10^{-3}$ meV/atom). Table II is a different boundary condition: matched-functional tetramer P changes from $\mathcal{S}_{\mathrm{rigid}}^{\mathrm{PBE+D3}} = +1.19$ meV/atom to $\mathcal{S}_{\mathrm{relaxed}} = -2.28$ meV/atom, so the sign is not preserved and $|\mathcal{S}|$ increases. That tetramer grid does not affine-scale the molecules and is not a retention ratio for the periodic scan. Table III records a tetramer seed diagnostic that is likewise not strain physics. The physically interpreted ionic-relaxation and placement tests are the periodic n=1 P four-corner set and the periodic n=4 three-seed set (seeds 42, 137, and 271).

C. Strain–dopant coupling energy

$\mathcal{S}_\delta(\epsilon)$ is the discrete strain–substitution interaction for a fixed dopant species $\delta \in \{\mathrm{B}, \mathrm{N}, \mathrm{P}\}$ at +3% load. It is not a continuum derivative $\partial^2 E/\partial\epsilon\,\partial\delta$, not an order parameter, and not interchangeable with a mixed derivative unless a local composition coordinate is introduced and higher-order remainders are negligible (Supplemental Material, Table II and Sec. II). B, N, and P are distinct chemical labels, not three numerical points on one continuous δ axis.

$$\mathcal{S}_\delta(\epsilon) = \Delta E(\epsilon, \delta) - \big[\Delta E(\epsilon, 0) + \Delta E(0, \delta)\big], \qquad (2)$$

with $\Delta E(\epsilon, \delta) = \big[E(\epsilon, \delta) - E(0, \mathrm{pristine})\big]/N_{\mathrm{atoms}}$ and δ=0 denoting the pristine reference at the same supercell size $n \times \mathrm{C}_{60}$. Equivalently, on corners $(0, 0)$, $(\Delta\epsilon, 0)$, $(0, \Delta\delta)$, $(\Delta\epsilon, \Delta\delta)$,

$$\mathcal{S}_\delta(\epsilon) = \frac{E(\epsilon, \delta) - E(\epsilon, 0) - E(0, \delta) + E(0, 0)}{N_{\mathrm{atoms}}}, \qquad (3)$$

which depends only on the four corner totals (reference independence). The four corner totals are evaluated on the same neutral Born–Oppenheimer surface; the stencil is not a charged-defect or polaron-rate estimator.

a. Why a four-point stencil? Eqs. (2)–(3) are the minimal rectangular stencil that uses four DFT totals, is second-order accurate in the bilinear term, cancels first-order increments and a shared offset, and equals the additive-screening bias by construction. Asymmetric three-point schemes retain linear contamination; other estimators are compared in Supplemental Material, Sec. II C.

b. Finite-difference link after a local composition coordinate. The species label δ is discrete. A mixed derivative $\partial^2 e/\partial\epsilon\,\partial x$ is defined only after a local continuous composition coordinate x is introduced at that fixed species (main text, Eq. (1)). Expanding each corner total in (ϵ, x) to second order and forming Eq. (3) cancels the first-order increments identically (Supplemental Material, Sec. II B), yielding

$$\mathcal{S}_\delta(\epsilon) \approx \frac{\partial^2 e}{\partial\epsilon\,\partial x}\,\Delta\epsilon\,\Delta x + \mathcal{O}(\Delta\epsilon^2, \Delta x^2), \qquad (4)$$

where Δx is a formal step of that local coordinate, not a spacing that places B, N, and P on one axis. Eq. (4) is therefore not $\partial^2 E/\partial\epsilon\,\partial\delta$ with B/N/P treated as numerical points of a shared δ. At finite +3% biaxial load and discrete B/N/P labels, higher-order terms in Eq. (4) are not separately resolved on the production periodic map; multi-amplitude strain audits (ϵ = 1–3%) are targeted sensitivity tests (Limitations). Throughout the main text, "nonlinear coupling" refers to a nonzero discrete strain–substitution interaction estimated by $\mathcal{S}_\delta(\epsilon)$, not to $\mathcal{S}$ itself as a fundamental constant and not to a continuum mixed derivative along a shared dopant axis.

$\mathcal{S}$ is reported in meV per atom and is invariant under a uniform energy offset shared by all four corners (Supplemental Material, Sec. II D).

c. Error propagation. If each corner carries uncorrelated equal per-atom uncertainty σ_e,

$$\sigma_{\mathcal{S}}^2 = 4\sigma_e^2, \qquad \sigma_{\mathcal{S}} = 2\sigma_e, \qquad (5)$$

from the four-term difference in Eq. (3) (derivation: Supplemental Material, Sec. II E). Table I lists contributions to the reporting band; we use $\sigma_{\mathcal{S}} \lesssim 2$ meV/atom as a conservative protocol floor on the $n \leq 4$ grid. $\mathcal{S} < 0$ (> 0) means the coupled corner is more (less) stable than the additive estimate. Additive-failure magnitude is

$$\eta \equiv \frac{|\mathcal{S}|}{|\Delta E_{\mathrm{coupled}}|}, \qquad (6)$$

where $\Delta E_{\mathrm{coupled}} = (E(\epsilon, \delta) - E(0, 0))/N_{\mathrm{atoms}}$.

D. Validation benchmarks

Interpretation rests on four checks: numerical hygiene (Table I); ionic relaxation of periodic n=1 P (Fig. 2(c)), where relaxation suppresses the constrained cross term; a separate tetramer boundary condition for P that does not affine-scale the molecules and reverses the sign while increasing $|\mathcal{S}|$ (Table II); and periodic n=4 placement seeds 42, 137, and 271 (Table IV). Table III is a cluster configuration diagnostic, not a strain grid. Periodic n=4 P relaxation, B/N periodic relaxations, dense-k / meta-GGA four-corner checks, and two-body image tests beyond n=8 are not reported (Table I; Limitations). A calculation-group inventory is given only in the Supplemental Material.

TABLE I. Error budget for four-corner $\mathcal{S}$ [Eq. (3)]. $\sigma_{\mathcal{S}} \approx 2\sigma_e$ when per-atom corner errors σ_e are uncorrelated and equal [Eq. (5)]; shared XC, k, and cell settings induce partial cancellation. Rows marked "not reported" are targeted sensitivity tests (Limitations); no meta-GGA or dense-k numbers appear without converged outputs. The reference-test rows use the fact that the pristine +3% strain cost per atom must be independent of supercell size: its deviation from the converged n=1, 4, 6 plateau is a measured systematic uncertainty on that row of $\mathcal{S}$, since $\mathcal{S}$ contains the pristine strain term with a negative sign. The ±2 meV/atom band is a conservative protocol floor for the converged rows, not a quadrature sum, and does not cover the two reference-limited sizes.

Source	$\lvert\Delta\mathcal{S}\rvert$ or $\sigma_{\mathcal{S}}$ (meV/atom)	Status
SCF (EPS_SCF = 10^{-6})	$\lesssim 2$ ($\sigma_{\mathcal{S}} \approx 2\sigma_e$)	Analytic bound
Pristine reference, n=1, 4, 6	≤ 0.06	Measured (reference test)
Pristine reference, n=2 (EPS_SCF = 10^{-5})	4.2	Measured (reference-limited row)
Pristine reference, n=8	5.4	Measured (reference-limited row)
Cutoff (350 vs 400 Ry)	< 0.2	Partial ($6 \times C_{60}$ N, +3%)
D3 cross term	$\lvert\mathcal{S}_{\mathrm{vdW}}\rvert \approx 0.33$	Tetramer P
D3 (B/N periodic)	—	Not reported
Ionic relaxation (periodic n=1 P; $\mathcal{S}_{\mathrm{rigid}} \to 0$)	$\gg 2$	Fig. 2(c)
k mesh (Γ vs $2 \times 2 \times 1$)	—	Not reported
XC (PBE+D3 vs SCAN / r^{2}SCAN)	—	Not reported
Adopted reporting band ($n \leq 4$): ±2 meV/atom.		

TABLE II. Invalid cluster/vacuum-box diagnostic (not strain physics). The tetramer four-corner grid does not affine-scale molecular coordinates; the tabulated +1.19 meV/atom rigid value and ≈1.9 matched-functional ratio therefore do not bound monolayer $\mathcal{S}$. The valid large-$\lvert\mathcal{S}\rvert$ test is periodic n=1 P (PBE+D3, fixed-cell): $\mathcal{S}_{\mathrm{rigid}} = -31.9$ meV/atom $\to \mathcal{S}_{\mathrm{relaxed}} \approx 0$. Numbers below are retained only as a protocol diagnostic.

Quantity	Rigid (PBE+D3 matched)	Relaxed (geometry opt.)
E(pristine, ϵ=0) (E_{h})	−1363.8769	−1363.9551
E(pristine, ϵ=3%) (E_{h})	−1363.8768	−1363.9550
E(P, ϵ=0) (E_{h})	−1370.8552	−1374.4415
E(P, ϵ=3%) (E_{h})	−1370.8446	−1374.4615
$\mathcal{S}$ (meV/atom)	+1.19	−2.28
Sign of $\mathcal{S}$ vs. rigid	+	− (not preserved)

TABLE III. Alternate-placement tetramer cluster map (PBE+D3 rigid; seed 137; six vacuum-box labels per dopant). Cartesian coordinates are *not* affinely scaled, so this table is a configuration diagnostic, not a strain-physics grid. Not used for main-text figures or for the quantitative $\mathcal{S}(n)$ claims; periodic $\mathcal{S}(n)$ is in Table IV. Periodic n=4 placement (seeds 42, 137, and 271) is the valid generality test (ranking $P{>}B{>}N$ in all three maps). The cluster $\lvert\alpha\rvert$ vs. $\lvert\mathcal{S}\rvert_{n=4}$ comparison in Fig. 3(d) is qualitative only. Linear-fit standard errors on six-point cluster maps are of order 10^2 meV/%, so α entries are not resolved monolayer strain coefficients.

Dopant	Placement	α (meV/%)	$\mathcal{S}^{\mathrm{tet}}_{+3\%}$ (meV/atom)
B	seed 137	−45	+1.9
N	seed 137	−60	+0.3
P	seed 137	+1028	−32.3

1. *Alternate-placement benchmark*

An independent dopant configuration at the same nominal ∼5 at.% loading repeats the B/N/P cluster map under PBE+D3 (Table III); that map does not affine-scale molecular coordinates. Main-text quantitative $\mathcal{S}(n)$ uses the periodic four-corner grid (Table IV); tetramer α and $\mathcal{S}^{\mathrm{tet}}_{+3\%}$ are cluster diagnostics and are not mixed with those periodic values. At periodic n=4, two independent site maps repeat the four-corner Eq. (2) audit; the $\lvert\mathcal{S}\rvert$ ranking at +3% matches the production map in Table IV.

2. *Ionic-relaxation benchmark*

Two boundary conditions are kept separate. The periodic n=1 P four-corner fixed-cell BFGS protocol (force thresholds 4.5×10^{-4} and 3.0×10^{-4} Ha/bohr, ∼0.023 and ∼0.015 eV/Å; displacement thresholds 3.0×10^{-3} and 1.5×10^{-3} bohr) is the constrained diagnostic reported in Fig. 2(c): ionic relaxation suppresses the cross term to the reporting floor. Rigid $\lvert\mathcal{S}\rvert$ on this path is an upper-end protocol estimate relative to that fully relaxed fixed-cell protocol, not a universal bound on every substrate, adhesion, or load path. The tetramer pristine/P four-corner BFGS grid does *not* affine-scale molecular coordinates (only the vacuum box changes). Table II is therefore a distinct boundary condition, not a retention ratio for the periodic series: the cross term reverses sign and increases in magnitude (+1.19 → −2.28 meV/atom).

These thresholds are tighter than typical 0.02 eV/Å defaults because relaxation can drive $|\mathcal{S}|$ toward zero and must be resolved at sub-meV/atom scale.

3. *Hirshfeld population along strain*

Hirshfeld atomic charges are computed along $\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$ on periodic n=1 B/N/P cells (spin-polarized KS for odd electron counts). They complement geometric $\bar{d}$ in Table V.

III. RESULTS

Total-energy audits use matched periodic supercells and the reference tetramer map (Sec. II C). Mechanism is deferred to Sec. IV.

A. Electronic structure under strain and doping

Figure 1(a) shows the reference tetramer: pristine and B/N/P-substituted cages at ϵ=0 with highlighted substituent sites (covalent-radius excess in Table V). Panels (b)–(d) are archived cluster-map / vacuum-box slices, not monolayer biaxial-strain screens: the π-DOS ridgelines of pristine, B, N, and P are stacked by the vacuum-box label; the frontier HOMO–LUMO separation $E_{\mathrm{H-L}}$ separates pristine, which retains 1.0–1.5 eV over the archived labels $\epsilon = -5$ to $+5\%$, from all three substituted cages, which close to $|E_{\mathrm{H-L}}| \lesssim 0.2$ eV and are mutually indistinguishable at the ϵ=0 snapshot within that window (+0.03, −0.14, and +0.05 eV for B, N, and P); panel (d) is the pristine cluster-map ΔE path, not a monolayer mechanical α. The negative entries mark inverted frontier-level ordering (gap closure), not a physical negative gap, and PBE underestimates gaps, so only the pristine-vs.-substituted contrast is interpreted. Supplemental Material gives full π-DOS stacks and periodic VBM/CBM isosurfaces (N size series; P n=4): frontier density remains cage-localized for n = 2–6.

B. Non-additive strain–doping coupling

Figure 2 is the central audit. Panel (d) defines the test: strain-only and substitution-only increments are summed and compared with the coupled four-corner total; $\mathcal{S}$ [Eq. (3)] is that signed gap. Panel (a) and Table IV report the fifteen periodic +3% $\mathcal{S}(n, \delta)$ values; panel (b) resolves each $\mathcal{S}$ into the strain cost on the pristine and on the substituted lattice. On the matched 400 Ry core (n=1, 2, and 4), P stays large and negative (31.9 meV/atom at n=1; 23.7 meV/atom at n=4) and B stays negative. Supercells with $n \geq 6$ (350 Ry) are reference-only. The apparent n dependence must, however, be read against a reference test that the four-corner construction makes available at no extra cost. The pristine +3% strain cost per atom is the same physical quantity in every supercell, so its spread across n measures protocol reproducibility directly. It is 2.99, 3.05, and 3.10 meV/atom at n=1, 4, and 6—reproducible to 0.06 meV/atom, two orders of magnitude tighter than the ±2 meV/atom reporting band—but −1.18 meV/atom at n=2 and 8.49 meV/atom at n=8. Because $\mathcal{S}$ contains the pristine strain term with a negative sign, those two rows inherit systematic offsets of 4.2 and 5.4 meV/atom (Table I); the n=2 cells are the ones that required a relaxed EPS_SCF = 10^{-5} (Sec. II). We do not rescale the n=2 or n=8 references onto the plateau to infer size independence, and we do not remove the n=8 sign change by that rescaling. N is small and positive only on the plateau-matched 400 Ry corners (n=1 and n=4); the n=2 and n=8 rows stay reference-limited and are not evidence of a size trend. At n=4, on the converged reference, $\mathcal{S} = -6.2 \pm 2$ (B), $+3.0 \pm 2$ (N), and -23.7 ± 2 meV/atom (P), where ± is the conservative protocol floor (Table I). The three substituents are therefore not equally resolved on the 400 Ry core. Referred to the protocol floor, $|\mathcal{S}|$ for P is 12–16 times the 2 meV/atom band at n=1 (31.9) and n=4 (23.7); N at n=1 (+2.30 meV/atom) lies at that floor and is resolution-limited. We treat $\mathcal{S}$ as numerically resolved only when its magnitude exceeds the conservative protocol uncertainty by a clear margin; values close to the floor are not assigned physical significance. Expressed as the additive-failure fraction of Eq. (6), η = 8–12% for P but stays below 1% for B (0.2–0.9%) and N (0.1–0.4%). Only P is thus large enough on this core grid to alter energy differences at the scale relevant to energetic screening when candidate separations are comparable to $|\mathcal{S}|$, B carries a smaller resolved cross term, and N sits at the detection floor—a stratification we treat as the main quantitative result rather than a uniform failure of additivity. The representative value on the matched core is n=4 ($|\mathcal{S}| = 23.7 \pm 2$ meV/atom, $\eta \approx 8\%$, about twelve

TABLE IV. Periodic coupling-energy audit at +3% biaxial strain (PBE+D3 rigid protocol, reference placement; core $n \leq 4$ at 400 Ry). $\mathcal{S}$ in meV/atom from Eq. (2); conservative protocol uncertainty ±2 meV/atom for $n \leq 4$ (Sec. II C). Substitutional loading is one dopant per C_{60} unit (∼1.67 at.% of total atoms in each supercell). Rows with $n \geq 6$ (350 Ry) are in Supplemental Material Table S1.

n	Dopant	ϵ (%)	at.%	$\mathcal{S}$ (meV/atom)
1	B	+3	1.67	−8.38
1	N	+3	1.67	+2.30
1	P	+3	1.67	−31.91
2	B	+3	1.67	−2.18
2	N	+3	1.67	+6.85
2	P	+3	1.67	−28.29
4	B	+3	1.67	−6.23
4	N	+3	1.67	+2.95
4	P	+3	1.67	−23.71

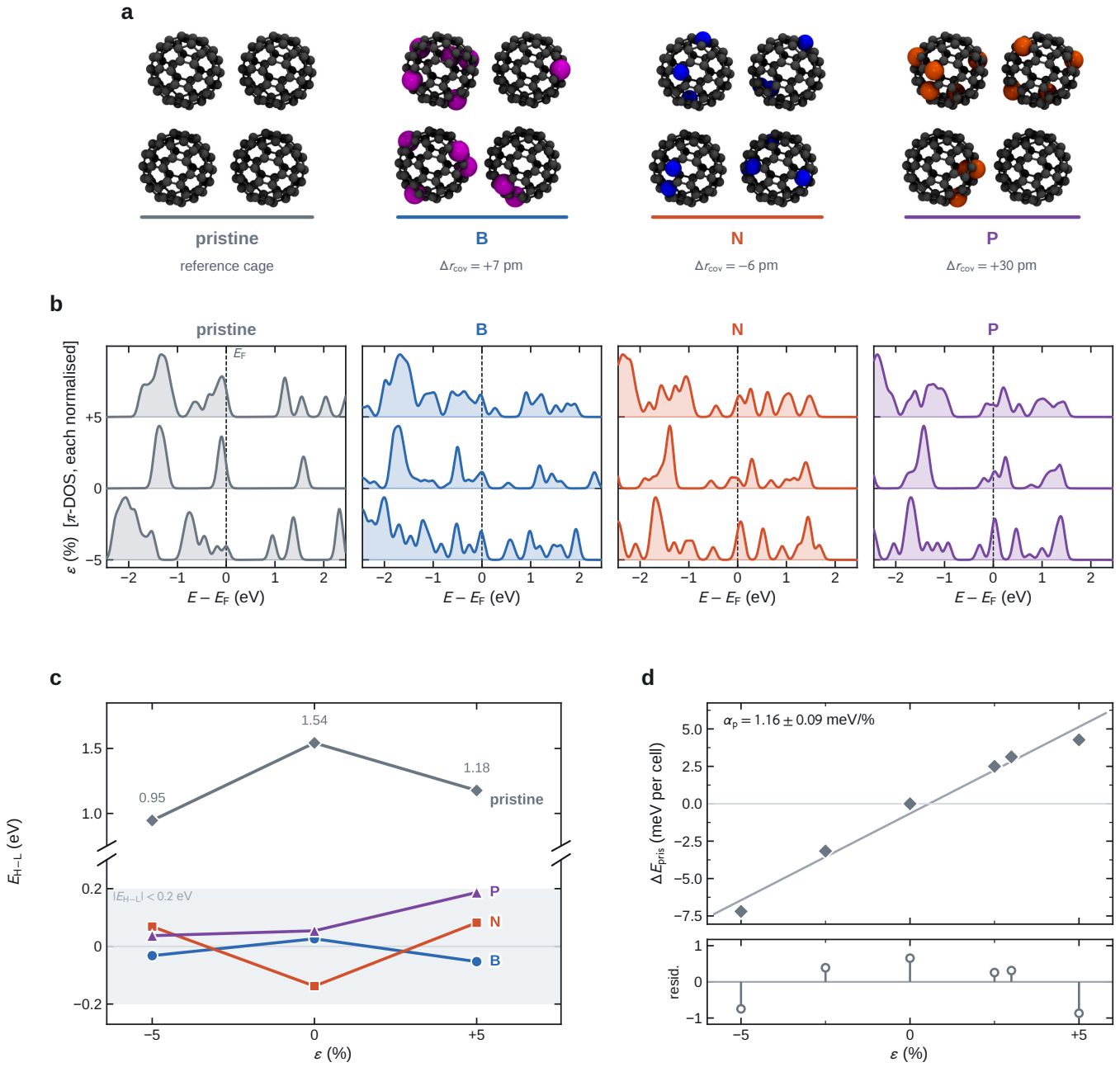


FIG. 1. Archived tetramer cluster map (PBE+D3). Cartesian coordinates are *not* affinely scaled; labeled ϵ points change only the vacuum box. (a) qHP C_{60} tetramers at the archived ϵ=0 snapshot: pristine and B/N/P-substituted cages, with the substituent sites highlighted (covalent-radius excess Δr_{cov} below each label; Table V). Heteroatom colors in the renders are CPK and differ from the dopant palette of the other panels. (b) π-DOS of the four systems at three archived vacuum-box labels ($\epsilon = -5, 0, +5\%$), each spectrum normalized to its own maximum (E_{F}=0). (c) Frontier HOMO–LUMO separation $E_{\mathrm{H-L}}$ from the archived π-DOS spectra, on a broken axis: pristine (upper segment) keeps a 1.0–1.5 eV gap, while all three substituted cages close it to $|E_{\mathrm{H-L}}| \lesssim 0.2$ eV (shaded band). The residual separation changes sign because the frontier levels cross—a gap-closure indicator, not a negative gap. PBE underestimates gaps, so only the pristine-vs.-substituted contrast is interpreted. These spectra are not monolayer biaxial-strain paths. (d) Pristine cluster-map energy ΔE_{pris} relative to ϵ=0 per 240-atom tetramer cell, with the least-squares line (slope $\alpha_{\mathrm{p}} = 1.16 \pm 0.09$ meV/%, standard error) and its residuals. This is a configuration diagnostic and not a monolayer mechanical α. Periodic joint-load four-corner audits are in Fig. 2.

times the 2 meV/atom reporting floor). Across the audited sizes the P cross term spans 23.7–31.9 meV/atom. The n=1 extremum (31.9±2 meV/atom) is the strongest constrained coupling on this grid and the weakest Γ-only support; it may include enhanced periodic-image coupling at the smallest cell and is not an intrinsic dilute-limit constant. We do not extrapolate $\mathcal{S}_\infty$. Rows with $n \geq 6$ (350 Ry) are in Supplemental Material Table S1 and are not used for mechanistic or asymptotic inference. Negative $\mathcal{S}$ (B, P at n=4) means the coupled corner lies below the additive estimate; positive $\mathcal{S}$ (N) means it lies above. Panel (c) compares the rigid protocol with the fully relaxed fixed-cell protocol studied here. Rigid $|\mathcal{S}|$ is an upper-end estimate for that pair of protocols, not a universal bound on every constraint manifold. On periodic n=1 P at +3%, ionic relaxation suppresses the constrained cross term from $\mathcal{S}_{\rm rig} \approx -31.9$ meV/atom to $|\mathcal{S}_{\rm relaxed}| < 10^{-3}$ meV/atom. Table II is a different boundary condition: matched-functional tetramer P changes from $\mathcal{S}^{\rm PBE+D3}_{\rm rigid} = +1.19$ meV/atom to $\mathcal{S}_{\rm relaxed} = -2.28$ meV/atom, so the sign is not preserved and $|\mathcal{S}|$ increases. That tetramer grid does not affine-scale the molecules and is not a retention ratio for the periodic scan. Relaxation can therefore reduce, reverse, or increase $|\mathcal{S}|$, depending on the structural branch and the boundary condition. Panel (e) is a schematic of the bilinear $c\,\epsilon x$ term at fixed dopant species after a local composition coordinate x is introduced, not a DFT fit and not a derivative along a shared δ axis.

C. Mechanical strain sensitivity

Pristine tetramers yield $\alpha_{\rm pris} = +1.2 \pm 0.1$ meV/% on the six-point reference grid. On the same cluster map, least-squares slopes versus the vacuum-box label are $+59 \pm 116$ (B), -297 ± 110 (N), and $+456 \pm 870$ meV/% (P) [Fig. 3(d)]. These numbers are cluster diagnostics, not monolayer strain coefficients. Only N is resolved beyond 2σ; the B and P slopes are not statistically distinguishable from zero, so α is reported as an order-of-magnitude descriptor and the strain response of the substituted tetramers is manifestly not well described by a single linear coefficient. That poor conditioning is itself part of the argument: a descriptor this noisy cannot be the basis for the additive extrapolation that sequential screens perform. Table V reports $\bar{d}$, $\sigma(\bar{d})$, and $\Delta r_{\rm cov}$. At +3%, P shows $\Delta\bar{d} \approx 3 \times 10^{-5}$ Å versus $\sim 8.5 \times 10^{-3}$ Å for B.

Figure 3 compares periodic Hirshfeld charges with the cluster-map geometry diagnostic. Hirshfeld $q(\epsilon)$ and $\bar{d}(\epsilon)$ [panels (a,b)] vary with dopant and strain. Local disorder $\sigma(\bar{d})(\epsilon)$ [panel (c)] broadens for B and N under tension while remaining near the ϵ=0 baseline for P. Panel (d) maps reference-map $|\alpha|$ against periodic $|\mathcal{S}|_{n=4}$ (marker area $\propto |\mathcal{S}|_{n=1}$): N lies in the high-$|\alpha|$/low-$|\mathcal{S}|_{n=4}$ quadrant and P in the high-$|\mathcal{S}|$ region—a qualitative rank inversion among three dopants (panel (e)), not a fitted correlation. P has the largest covalent-radius mismatch ($\Delta r_{\rm cov} = +30$ pm; Table V) and the largest audited $|\mathcal{S}|$ under the rigid protocol.

IV. DISCUSSION

A. Main finding

Additive screening omits the four-corner cross term $\mathcal{S}$ under constrained joint load. On the matched 400 Ry core the representative value is n=4 ($|\mathcal{S}| = 23.7 \pm 2$ meV/atom). That cross term is large enough to alter energy differences at the scale relevant to energetic screening for P when candidate separations are comparable to $|\mathcal{S}|$ (Table IV), while linear α does not rank $|\mathcal{S}|$ [Fig. 3(d)]. The three-dopant pattern is selective and illustrative: only P is resolved well above both the protocol floor (∼2 meV/atom) and the $\eta > \sim$1% flag, B clears the floor but not the η flag, and N at n=1 (+2.30 meV/atom) is resolution-limited (Sec. III B). That selectivity is consistent with a larger covalent-radius mismatch for P; the present data do not identify a unique cause. Ionic relaxation of periodic n=1 P suppresses the constrained cross term ($|\mathcal{S}_{\rm relaxed}| < 10^{-3}$ meV/atom). On the tetramer boundary condition the same relaxation reverses the sign and increases $|\mathcal{S}|$ (+1.19 to −2.28 meV/atom; Table II; Fig. 2(c)). The pattern is consistent with a geometry-mediated contribution; the data do not partition elastic and electronic energy (Sec. IV E). We treat $\mathcal{S}$ as numerically resolved only when $|\mathcal{S}|$ exceeds the protocol floor by a clear margin. The $\eta > \sim$1% flag is a practical marker for coupled re-evaluation on this PBE+D3 map, not a 95% confidence interval and not a thermodynamic boundary.

B. Mechanism

For size-mismatched P, the correlated pattern is (Fig. 4): $\Delta r_{\rm cov} = +30$ pm together with the largest $|\mathcal{S}|$. This is a consistency check on the present energetic and structural audits, not a partitioned causal chain. Two features of the audited set are consistent with a geometry-mediated contribution, not a unique control (Table V; Fig. 4(b,d)). First, the *sign* tracks the sign of the mismatch: N is the only undersized substituent ($\Delta r_{\rm cov} = -6$ pm) and the only one with $\mathcal{S} > 0$, while the oversized B and P (+7 and +30 pm) both give $\mathcal{S} < 0$. Second, the *rank* of the magnitude follows $|\Delta r_{\rm cov}|$ at both audited sizes: N < B < P for $|\Delta r_{\rm cov}|$ ($6 < 7 < 30$ pm) and for $|\mathcal{S}|$ at n=1 ($2.3 < 8.4 < 31.9$ meV/atom) and n=4 ($3.0 < 6.2 < 23.7$ meV/atom). Radius nonetheless fails to fix the proportionality constant: $|\mathcal{S}|_{n=1}/|\Delta r_{\rm cov}|$ is ∼0.4 meV/atom per pm for N against ∼1.1–1.2 for B and P, so a single geometric descriptor orders the three substituents without predicting their magnitudes. With three substitutional labels this is a consistency check

TABLE V. Local dopant–cage structure from archived *tetramer* XYZ (∼5% nominal doping; qualitative proxy for mismatch trends). Cartesian coordinates are *not* affinely scaled across labeled ϵ snapshots. Mean nearest-neighbor C distance $\bar{d}$ in Å; standard deviation $\sigma(\bar{d})$ across dopant sites; covalent-radius excess vs. C in pm. The tabulated $\Delta\bar{d} = \bar{d}(\epsilon{=}+3\%) - \bar{d}(\epsilon{=}0)$ is a frozen-geometry cluster diagnostic, not a monolayer nearest-neighbor stretch. Distances are DFT-derived from archived tetramer XYZ; no experimental bond errors are quoted—comparison is internal across dopants and with qHP literature on intrinsic cage strain [10, 14].

Dopant	$\bar{d}(\epsilon{=}0)$ (Å)	$\bar{d}(\epsilon{=}+3\%)$ (Å)	$\Delta\bar{d}$ (Å)	$\sigma(\bar{d})$ @ +3% (Å)	Δr_{cov} (pm)
B	1.384	1.393	+0.0085	0.019	+7
N	1.393	1.402	+0.0087	0.024	−6
P	1.384	1.385	+0.00003	0.0004	+30

on the misfit picture, not a fitted scaling law. Electronic redistribution does not reproduce either pattern: the fractional Hirshfeld change $|\Delta q/q|$ over $0 \to +3\%$ is largest for N (∼18%), intermediate for P (∼10%), and smallest for B (∼6%), which neither orders nor signs $\mathcal{S}$ [Fig. 4(a)]. π-DOS morphology likewise does not order $|\mathcal{S}|$: the substituted cages are indistinguishable in $E_{\mathrm{H-L}}$ at $\epsilon{=}0$ yet $|\mathcal{S}|$ at $n{=}1$ runs from 2.3 to 31.9 meV/atom [Figs. 1(c) and 3(d)]. Uniform bond stretch does not track $|\mathcal{S}|$: B and N show comparable $\Delta\bar{d}$ at +3% ($+8.5 \times 10^{-3}$ and $+8.7 \times 10^{-3}$ Å) yet $|\mathcal{S}|$ of intermediate and smallest size, whereas P barely stretches ($+3 \times 10^{-5}$ Å) and couples most strongly. Ionic relaxation of periodic $n{=}1$ P suppresses the constrained cross term; on the tetramer the same relaxation reverses the sign and increases $|\mathcal{S}|$ (Fig. 2(c)). Grimme D3 contributes $\mathcal{S}_{\mathrm{vdW}} \approx +0.33$ meV/atom at tetramer P, about 28% of the small tetramer $|\mathcal{S}| \approx 1.2$ meV/atom (Table I). That decomposition does not indicate that dispersion dominates the larger periodic P signals (23.7 and 31.9 meV/atom).

C. Relation to literature

This is a periodic total-energy audit of joint (ϵ, δ) load (Table IV), not a strain-only, doping-only, or mobility/gap map [3, 4, 9, 12, 13]. Strain-only electronic and optical maps of C_{24} graphullerene [17] likewise omit the matched four-corner $\mathcal{S}$ audit used here for qHP C_{60}. Cluster expansions already encode $c\,\epsilon x$ [5, 6]; the increment is the matched four-corner audit on a soft, pre-strained network with rigid-vs.-relaxed and placement checks (Tables II–IV; Supplemental Material, Table I). Prior graphullerene studies often relaxed ions first or added separate grids [4, 9, 12], which masks constrained-load coupling. Quasi-harmonic thermal-expansion maps of related fullerene networks [22] likewise omit a matched four-corner $\mathcal{S}$ audit under a frozen biaxial cell. Strain-induced band-alignment and mobility maps of 1D/2D C_{60} nanostructures [23] likewise omit that matched four-corner $\mathcal{S}$ audit. Endohedral doping of the same qHP cages [24] maps electronic and optical responses of encapsulated guests, not the substitutional B/N/P plus frozen-biaxial four-corner $\mathcal{S}$ protocol used here. Two-dimensional C_{70} networks have been proposed as synthesizable platforms for electronic band engineering [25]; that design axis is cage-allotrope and band-edge selection, not the substitutional B/N/P plus frozen-biaxial four-corner $\mathcal{S}$ audit used here for qHP C_{60}. Strain-induced magnetism and half-metallicity in C_{20} graphullerene monolayers [26] likewise address a magnetic/half-metallic response of a different cage allotrope, not the substitutional B/N/P plus frozen-biaxial four-corner $\mathcal{S}$ audit used here for qHP C_{60}. The four-corner $\mathcal{S}$ is a total-energy cross term on the *neutral* Born–Oppenheimer surface; charged versus neutral DFT formulations that are equivalent for polaron self-interaction corrections [27] do not convert that audit into a Marcus rate or a charged-defect formation energy.

D. Applicability and generality

The four-corner audit applies when local substitutional strain cannot fully relax on a short-range, elastically soft covalent network [10, 28]. It does not apply to fully relaxed bulk references, metallic or liquid hosts, or high-T regimes where configurational entropy dominates sub-meV differences (Sec. IV E). Uniaxial load should preserve the stencil but rescale c through elastic anisotropy; that case is not audited.

Laboratory tests need not resolve $\mathcal{S}$: cage-mode Raman/IR or substrate-bending that deviate from separate strain-plus-doping calibrations would signal additive failure, provided chemistry and strain are matched and the comparison is to a constrained, not fully relaxed, state.

E. Limitations

Main-text α and $\mathcal{S}$ use PBE+D3. Periodic production cells are affinely strained at fixed fractional coordinates; cluster tetramer grids change only the vacuum box and are not used as strain physics. Rigid $|\mathcal{S}|$ is an upper-end protocol estimate relative to the fully relaxed fixed-cell protocol studied here, not a universal bound on every constraint manifold. Ionic relaxation of periodic $n{=}1$ P suppresses the constrained cross term ($|\mathcal{S}_{\mathrm{relaxed}}| < 10^{-3}$ meV/atom). Table II is a different boundary con-

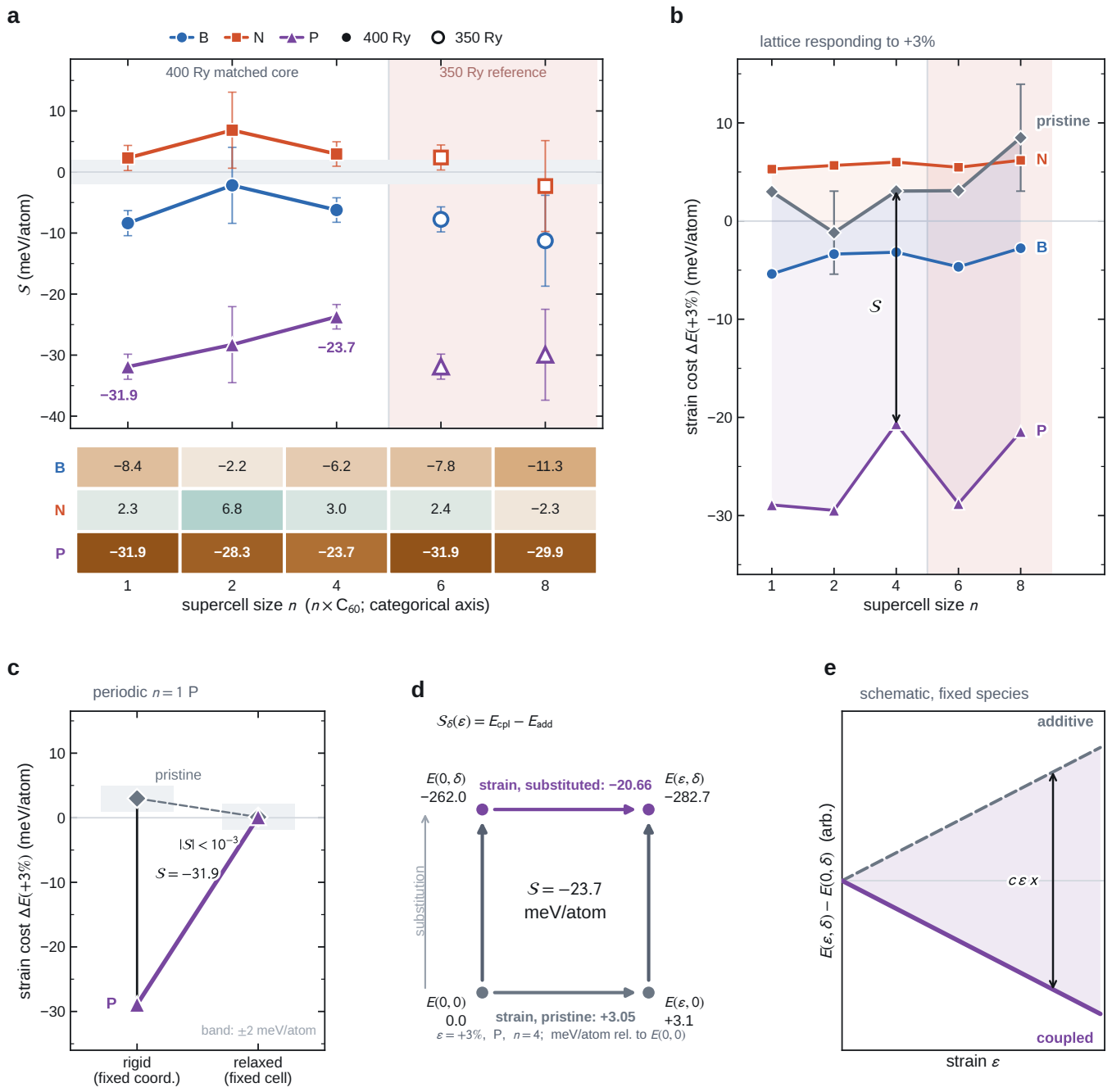


FIG. 2. Additive strain–dopant screening fails under constrained load. (a) Periodic +3% $\mathcal{S}(n, \delta)$ on a categorical n axis (gray band: ±2 meV/atom protocol resolution floor, not a statistical confidence interval). Filled markers and connecting lines are the matched 400 Ry core (n=1, 2, 4). Open markers are n=6, 8 at 350 Ry, shown as reference only and not used for mechanistic or asymptotic inference. Error bars add the systematic offset of the pristine strain reference; the larger n=2 bar is that offset on the 400 Ry core, not a cutoff-tier marker (Table I). Pink columns mark only the 350 Ry sizes. The strip below tabulates the same values (umber: $\mathcal{S} < 0$; teal: $\mathcal{S} > 0$; hatched cells lie within the floor). On the matched core, headline n=4 P is −23.7 meV/atom and the span end n=1 is −31.9 meV/atom, while several B/N points sit near the floor, so the audit resolves a stratification and not a uniform failure of additivity. (b) Origin of $\mathcal{S}$: the energy cost of +3% strain on the pristine lattice (gray; bars: systematic offset of the pristine reference) and on each substituted lattice (colors), per atom. The vertical gap between a colored and the gray curve is $\mathcal{S}$ (shaded). (c) Constrained fixed-coordinate diagnostic: an upper-end protocol estimate relative to the fully relaxed fixed-cell protocol studied here, not a universal bound on every constraint and not a second size-scaling panel. Periodic n=1 P at +3%: the strain cost on the pristine and P lattices differs by $\mathcal{S} = -31.9$ meV/atom at fixed coordinates, and ionic relaxation collapses the two to $|\mathcal{S}| < 10^{-3}$ meV/atom, below the reporting floor (shaded windows: ±2 meV/atom about the pristine value). Table II is a distinct tetramer boundary condition (vacuum box; coordinates not affinely strained): +1.19 → −2.28 meV/atom (sign reversal and larger magnitude). It is not a retention ratio for the periodic grid. (d) Definition at fixed dopant species δ: $\mathcal{S}_\delta(\epsilon) = E_{\rm cpl} - E_{\rm add}$ [Eq. (3)], evaluated for P at n=4 and ϵ=+3%. Corner energies are relative to $E(0, 0)$ (meV/atom); the strain costs on the pristine (+3.05) and substituted (−20.66) lattices differ by $\mathcal{S} = -23.7$ meV/atom. (e) Schematic at fixed species: the bilinear $c\,\epsilon x$ term, with x a local composition coordinate, separates the coupled path from the additive sum (cartoon in arbitrary units, not a DFT fit).

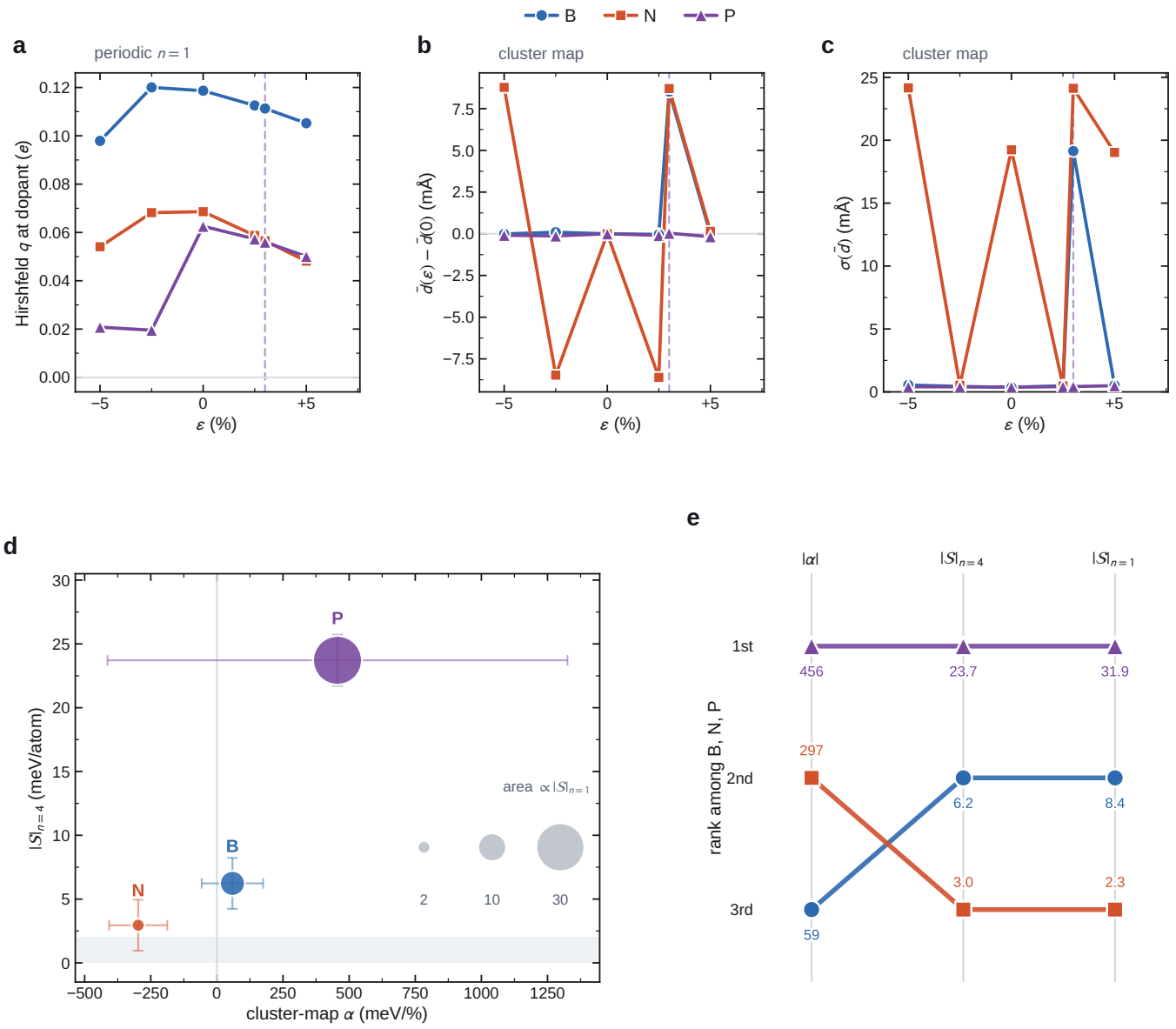


FIG. 3. Charge vs. geometry diagnostics and a qualitative rank comparison (n=3 dopants). (a) Periodic Hirshfeld $q(\epsilon)$ for B/N/P on affinely strained n=1 cells (six-point paths; dotted guide at +3%). This panel is a valid strain-path charge diagnostic. (b) Change of the nearest-neighbor distance $\bar{d}(\epsilon) - \bar{d}(0)$ vs. vacuum-box label from archived tetramer XYZ ($\bar{d}(0) = 1.384$, 1.393, 1.384 Å for B, N, P). The labels are not an affine monolayer strain path, and jumps between neighboring labels are not interpreted as a strain response. (c) Local disorder $\sigma(\bar{d})$ on the same frozen cluster map. (d) Signed cluster-map α vs. periodic $|\mathcal{S}|_{n=4}$ (marker area $\propto |\mathcal{S}|_{n=1}$; horizontal bars: linear-fit standard errors on the six-point cluster map; gray band: ±2 meV/atom floor). The cluster α is a configuration diagnostic, not a monolayer strain coefficient. (e) Rank of B, N, P by $|\alpha|$, $|\mathcal{S}|_{n=4}$, and $|\mathcal{S}|_{n=1}$ (values in the units of the other panels). N inverts a naive $|\alpha| \to |\mathcal{S}|$ expectation; with only three dopants panels (d) and (e) report qualitative ranking, not Pearson, R^2, or Spearman statistics.

dition: the tetramer P cross term reverses sign and increases in magnitude (+1.19 to −2.28 meV/atom) and is not a retention ratio for the periodic scan. Eq. (1) and the four-corner link [Eq. (4)] assume small perturbations on a single smooth energy branch; they lose validity at large strain (e.g. $\gtrsim$10%) or across structural phase transitions, and the production audits intentionally remain at +3% biaxial load inside the nominally elastic regime. Finer meshes at ±1, ±2, and ±4%, including $\mathcal{S}_P(1\%, 2\%, 3\%)$ on periodic n=1, remain outside the production audit. All totals are 0 K Born–Oppenheimer energies; finite-temperature free energies and configurational entropy remain unexplored and could reorder sub-meV competitors without restoring additivity if bilinear coupling persists at the electronic-structure level. Production periodic cells use Γ-only sampling; dense-k and SCAN/r^{2}SCAN

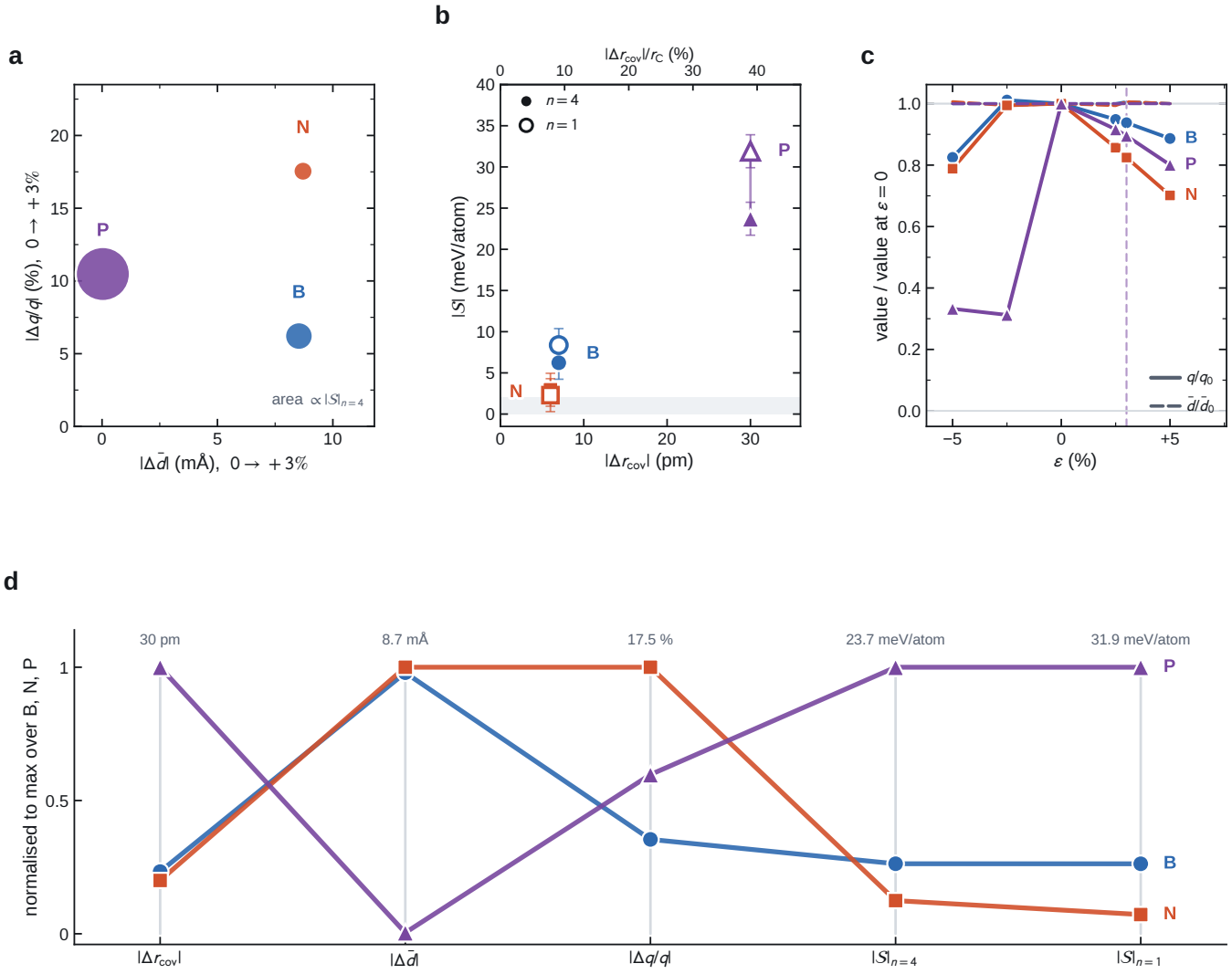


FIG. 4. Schematic mechanism consistent with the present energetic and structural audits (misfit strain, charge redistribution, and π-DOS morphology are correlated observables, not a partitioned causal chain). (a) Response plane from $0 \rightarrow +3\%$ path endpoints; bubble area $\propto |\mathcal{S}|_{n=4}$. Neither axis orders the coupling: B and N stretch comparably ($|\Delta\bar{d}| \approx 8.5$ and 8.7 mÅ) at intermediate and *smallest* $|\mathcal{S}|$, N also redistributes the most charge ($|\Delta q/q| \approx 18\%$), and P couples most strongly while barely stretching. (b) $|\Delta r_{\mathrm{cov}}|$ (top axis: relative to the C covalent radius) vs. $|\mathcal{S}|$ at n=4 (filled) and n=1 (open); bars and band: ±2 meV/atom floor. The rank follows mismatch (N < B < P) at both sizes, but the three points are not proportional, so radius orders without predicting magnitude. In this three-substituent audit the clearly large deviation is the largest mismatch (+30 pm, P). The panel does not mark a fitted threshold below which additivity holds. (c) Linked normalized paths $q(\epsilon)/q(0)$ (solid) and $\bar{d}(\epsilon)/\bar{d}(0)$ (dashed). (d) Parallel-coordinate summary: each observable is normalized to its maximum over B, N, P (maximum printed above each axis). P tops the mismatch and both $|\mathcal{S}|$ axes and is lowest in $|\Delta\bar{d}|$; N tops $|\Delta\bar{d}|$ (marginally above B) and $|\Delta q/q|$ and is lowest in $|\mathcal{S}|$. With three substituents these panels are consistency checks on the misfit picture, not fitted correlations.

four-corner checks are not reported (Table I). On the tested production grid, n=1 is the strongest constrained periodic coupling observed; it is not an intrinsic dilute-limit constant. That cell is also the one for which Γ-only sampling is least adequate, so the n=1 magnitude carries the weakest k-sampling support in the set. The present data do not establish size independence of $\mathcal{S}$. Two of the five supercell sizes (n=2, 8) have pristine references off the converged plateau by 4–5 meV/atom, so their $\mathcal{S}$ rows are reported but not used for size-trend claims; recomputing those four corners at EPS_SCF = 10^{-6} and matched cutoff is the cheapest remaining improvement to the audit. The n=8 offset is not a cutoff-class change: that supercell shares the n=6 production cutoff (Methods). Cells with $n \geq 6$ use 350 Ry and remain Supplemental reference rows; they are not used for mechanistic or asymptotic inference. Alternate placements (Table III) bound configuration specificity; Table V geometry is a

tetramer proxy (∼5 at.%) without Mayer/Bader partitioning or matched periodic n=4 bond statistics. With only three substitutional labels, the α–$|\mathcal{S}|$ map [Fig. 3(d)] is a qualitative rank comparison among n=3 chemistries, not a fitted correlation law or predictive model for linear α. Direct elastic-vs.-electronic energetic decomposition (Mayer bond order, Bader strain partitioning, virial stress) and dispersion cross-term audits for B/N periodic cells remain open (Table I).

F. Constrained load and outlook

Rigid $|\mathcal{S}|$ is an upper-end protocol estimate, relative to the fully relaxed fixed-cell protocol studied here, for epitaxial clamp, adhesion, or ultrafast load where ions cannot relax (Fig. 2(c)). It is not a universal bound on every constraint manifold and not an equilibrium free-energy difference. Ionic relaxation of periodic n=1 P suppresses the constrained cross term. On the tetramer P grid (Table II) the same relaxation reverses the sign and increases $|\mathcal{S}|$ (+1.19 to −2.28 meV/atom). For a new host: fix or relax coordinates to the loading scenario, evaluate four-corner $\mathcal{S}$, and replace sequential screening when $|\mathcal{S}|$ is numerically resolved well above the protocol floor (Fig. 2(d,e)). Larger-radius group-V substituents (As, Sb) are expected to couple more strongly than P under constrained load; that extrapolation is unverified. Follow-up: Mayer/Bader partitioning, $\mathcal{S}$(cutoff) and $\mathcal{S}(k)$, and periodic n=4 P relaxation.

V. CONCLUSIONS

Additive screening omits a four-corner cross term under constrained load on qHP C_{60}. On the matched 400 Ry core the representative value is n=4 ($|\mathcal{S}| = 23.7 \pm 2$ meV/atom, $\eta \approx 8\%$). The audited P span is 23.7–31.9 meV/atom; n=1 (31.9 ± 2 meV/atom) is the strongest constrained coupling and the weakest Γ-only support. B carries a smaller resolved cross term, and N sits near the reporting floor. $\mathcal{S}$ is a boundary-condition-dependent diagnostic, not an equilibrium universal material constant. The n=2 $\mathcal{S}(n)$ row is reference-limited at the undoped corner (Methods; Table I), not a sign change of the discrete species interaction. $\mathcal{S}_\delta(\epsilon)$ is the discrete strain–substitution interaction at fixed dopant species; a mixed derivative is defined only after a local composition coordinate is introduced. It is not a new thermodynamic observable. The four-corner $\mathcal{S}$ is a cross term among those neutral totals, not a charged-defect formation energy or a Marcus rate. Linear α does not rank $|\mathcal{S}|$ among B/N/P and is itself poorly conditioned on six-point tetramer grids (standard errors comparable to or exceeding the slopes), so it cannot license the additive extrapolation [Fig. 3(d)]. The larger $|\mathcal{S}|$ for P is consistent with a pre-strain, geometry-mediated contribution; the data do not uniquely partition elastic and electronic energy. Relaxation suppresses the constrained cross term on periodic n=1 P and reverses and increases $|\mathcal{S}|$ on the tetramer (Fig. 2(c)). In the present three-substituent audit the clearly large deviation occurs for the largest mismatch (+30 pm, P). That pattern is a consistency check on the misfit picture, not a 20 pm threshold and not a fitted scaling law. For the smaller mismatches audited here, coupled re-evaluation is optional, since B stays below $\eta \approx 1\%$ and N sits at the reporting floor. Other networks require their own four-corner audit before either conclusion is transferred. That same cage-allotrope caveat applies to C_{70} network band-edge proposals [25] and to strain-induced magnetism in C_{20} graphullerene [26]: neither replaces a four-corner $\mathcal{S}$ audit on the network under study.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

Computational inputs and outputs, workflow archives, tabulated coupling energies, four-corner total-energy components underlying $\mathcal{S}$, and materials to reproduce the figures and tables are archived at `https://github.com/chenxingqiang/sci-simukit` (branch `cursor/prb-manuscript-rigor-audit-1d10`, commit `230e279c4821d64e8db23987354c425ac720d869`). Ionic-relaxation, Hirshfeld, and size-scaling archives referenced in the main-text tables are included in that archive.

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